

JEE MAIN REVISION MASTER: BASICS OF ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 295 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	183 PYQs	Applied Topics
Very High	Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	95 PYQs	Quantitative Analysis
Very High	Basics Of Organic Chemistry — Core Principles & Governing Laws	44 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	$\frac{3.0\text{cm}}{5.0\text{cm}}=0.6, \quad \text{CH}_3\text{COOH} > \text{C}_6\text{H}_5\text{OH} > \text{H}_2\text{O} > \text{CH}_3\text{OH}, \quad \text{NHCOCH}_3$
Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	$-\text{OCH}_3, -\text{NO}_2, -\text{CN}, -\text{CH}_3-\text{NHCOCH}_3, -\text{COR}, -\text{OH}, -\text{COOH}, -\text{Cl}, \quad \text{MF}=\text{C}_9\text{H}_{12}, \quad (\text{CO}_3)^{2-}$
Basics Of Organic Chemistry — Core Principles & Governing Laws	$-\text{NO}_2, \quad \text{rightarrow sp}^2, \quad \text{c}(-\text{NO}_2) > \text{a}(-\text{CN}) > \text{b}(-\text{COOH}) > \text{d}(-\text{I})$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basics Of Organic Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 183 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Reaction List-I Mechanism A. Williamson Synthesis I. Electrophilic addition B. Friedel Craft R..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The strongest conjugate acid will result from :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Sublimation is used for the separation and purification of compounds with low melting point. ..." [Ans: A]

3.2 Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 95 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Vapours of the liquid with higher boiling point condense before vapours of the liquid with lo..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "For the given molecule, "x", the preferred site for the attack of the electrophile is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : There are several conformers for n-butane. Out of those conformers, Statement II : As the di..." [Ans: N/A]

3.3 Basics Of Organic Chemistry — Core Principles & Governing Laws (Very High Yield — 44 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Basics Of Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Basics Of Organic Chemistry PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Sublimation is used for the separation and purification of compounds with low melting point. ..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "Increasing order of electron withdrawing power of following functional groups is : a. -CN b. -COOH c. -NO₂ d. -I..." [Ans: N/A]
- [2026 Evening Shift] Q#3 (MCQ): "Shown below is the structure of methyl acetate with three different α, β and γ carbon - oxygen bonds. The correct order of ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 23rd January Morning Shift (Q#1)

Given below are two statements : Statement I : Sublimation is used for the separation and purification of compounds with low melting point. Statement II : The boiling point of a liquid increases as the external pressure is reduced. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are false **[Correct]**
- (B) Statement I is true but Statement II is false
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Correct Answer: A

Solution Analysis: Statement I: False. Sublimation is used to separate/purify sublimable (volatile) solids —solids that pass directly from solid to vapour on heating. This depends on having sufficient vapour pressure in the solid state, not simply on having a low melting point . Statement II: False. The boiling point decreases when external pressure is reduced (principle of vacuum distillation). ■ Correct option: A (Both Statement I and Statement II are false).

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

Identify the correct statements : The presence of $-\text{NO}_2$ group in benzene ring A. activates the ring towards electrophilic substitutions. B. deactivates the ring towards electrophilic substitutions. C. activates the ring towards nucleophilic substitutions. D. deactivates the ring towards nucleophilic substitutions. Choose the correct answer from the options given below :

- (A) and D Only
- (B) and D Only **[Correct]**
- (C) and A Only
- (D) and C Only

Correct Answer: B

Solution Analysis: Presence of NO_2 group on the benzene ring deactivates the ring towards electrophilic substitution reactions because it shows -M effect (it withdraws electrons from the ring by resonance). Because the NO_2 group pulls electron density away, the ring becomes less reactive towards electrophiles. At the same time, the NO_2 group activates the ring towards nucleophilic substitution because the withdrawal of electrons makes the ring more suitable for attack by nucleophiles (electron-rich species). Ans. $\rightarrow$ B & C

Benchmark Problem 3: JEE Main 2026 (Online) 28th January Morning Shift (Q#4)

CORRECT order of stability for the following is $\text{CH}_2=\text{CH}^-$, $\text{CH}_3-\text{CH}_2^-$, $\text{CH} \equiv \text{C}^-$

- (A) $\text{CH} \equiv \text{C}^- > \text{CH}_3-\text{CH}_2^- > \text{CH}_2=\text{CH}^-$
- (B) $\text{CH}_2=\text{CH}^- > \text{CH} \equiv \text{C}^- > \text{CH}_3-\text{CH}_2^-$
- (C) $\text{CH} \equiv \text{C}^- > \text{CH}_2=\text{CH}^- > \text{CH}_3-\text{CH}_2^-$ **[Correct]**
- (D) $\text{CH}_3-\text{CH}_2^- > \text{CH}_2=\text{CH}^- > \text{CH} \equiv \text{C}^-$

Correct Answer: C

Solution Analysis: Stability of carbanions depends mainly on how well the carbon can hold the negative charge. Step 1: Use NCERT idea — effect of hybridisation (electronegativity) Greater s-character $\Rightarrow$ carbon is more electronegative $\Rightarrow$ negative charge is more stable. sp carbon: 50% s-character (most electronegative) sp^2 carbon: 33% s-character sp^3 carbon: 25% s-character (least electronegative) So stability order due to hybridisation is: $\text{sp} > \text{sp}^2 > \text{sp}^3$ Step 2: Apply to given carbanions $\text{CH} \equiv \text{C}^-$ is on an sp carbon $\Rightarrow$ most stable $\text{CH}_2=\text{CH}^-$